

Foundation Mathematics 1017SCG
Week 8 Workshop Questions

1. Sketch a reasonably accurate graph of $y = |x|$.
2. Sketch each of the following using transformations of the graph $y = |x|$.

Hint: Some of the graphs will require multiple transformations.

- (a) $y = |x + 2|$
- (b) $y = |x| - 4$
- (c) $y = |x + 3| - 5$
- (d) $y = |x - 4| + 1$
- (e) $y = -|x| + 2$
- (f) $y = -|x - 1| + 3$

3. Consider the following piecewise function.

$$f(x) = \begin{cases} 2x - 1 & x \geq 1 \\ -x + 4 & x < 1 \end{cases}$$

- (a) Evaluate $f(-1)$.
- (b) Evaluate $f(0)$.
- (c) Evaluate $f(1)$.
- (d) Evaluate $f(2)$.

4. Consider the following piecewise function.

$$g(x) = \begin{cases} 2 - x & x \leq -2 \\ 5x + 1 & -2 < x < 3 \\ 2x - 4 & x \geq 3 \end{cases}$$

- (a) Evaluate $g(-3)$.
- (b) Evaluate $g(-2)$.
- (c) Evaluate $g(0)$.
- (d) Evaluate $g(5)$.

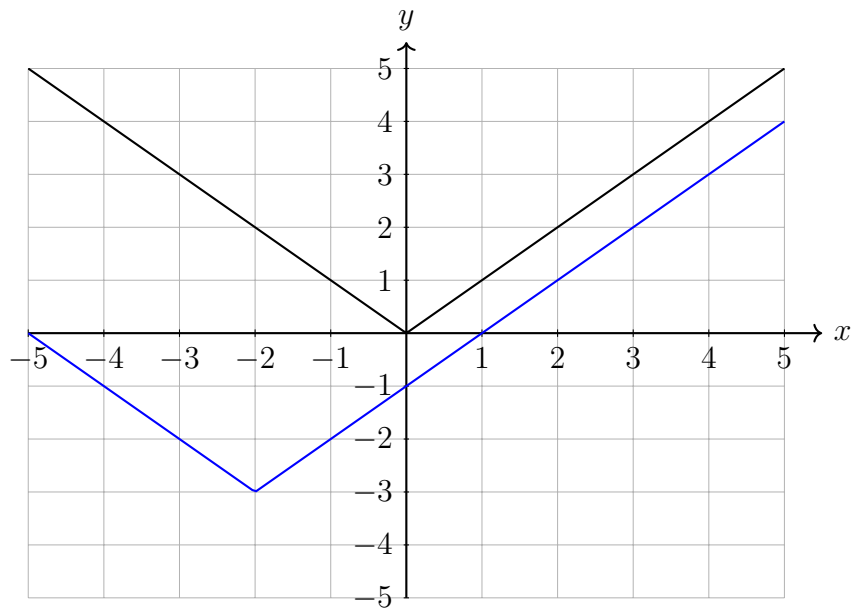
5. Consider the following piecewise function.

$$h(x) = \begin{cases} x^2 + 4x & x < -5 \\ 4 & -5 \leq x < 2 \\ \sqrt{x} & x \geq 2 \end{cases}$$

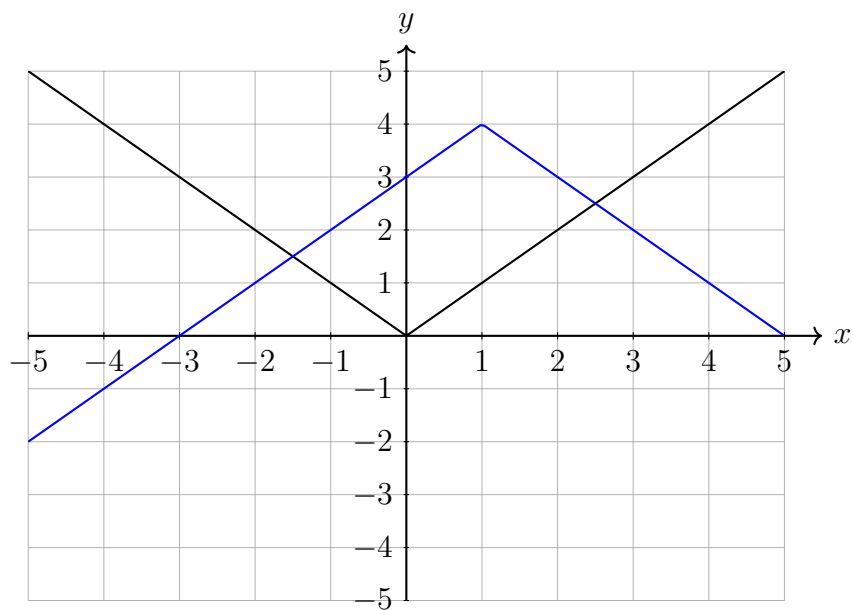
- (a) Evaluate $h(-3)$.
- (b) Evaluate $h(-6)$.
- (c) Evaluate $h(0)$.
- (d) Evaluate $h(9)$.

6. For each of the following functions, find its inverse function $f^{-1}(x)$.
- (a) $f(x) = 7x - 2$
 - (b) $f(x) = 3x + 4$
 - (c) $f(x) = \frac{2x - 1}{5}$
 - (d) $f(x) = \frac{6x - 7}{2}$
 - (e) $f(x) = 4e^{5x}$
 - (f) $f(x) = 7e^{-x}$
 - (g) $f(x) = 2e^{x-1}$
 - (h) $f(x) = e^{3x+4}$
7. Consider the functions $f(x) = 5x$ and $g(x) = 2x + 1$.
- (a) Find $f(g(x))$ and expand your answer. (This could also be written as $f \circ g$)
 - (b) Find $g(f(x))$ and expand your answer. (This could also be written as $g \circ f$)
8. Consider the functions $f(x) = 3x - 4$ and $g(x) = 7x + 2$.
- (a) Find $f(g(x))$ and expand your answer. (This could also be written as $f \circ g$)
 - (b) Find $g(f(x))$ and expand your answer. (This could also be written as $g \circ f$)
9. Consider the functions $f(x) = 4x$ and $g(x) = 2\cos(x)$.
- (a) Find $f(g(x))$. (This could also be written as $f \circ g$)
 - (b) Find $g(f(x))$. (This could also be written as $g \circ f$)
10. Consider the functions $f(x) = 3x - 1$ and $g(x) = 5\sin(2x)$.
- (a) Find $f(g(x))$. (This could also be written as $f \circ g$)
 - (b) Find $g(f(x))$. (This could also be written as $g \circ f$)

11. The graph of $y = |x|$ is shown below in black. Find an equation for the blue graph that uses the absolute value function.

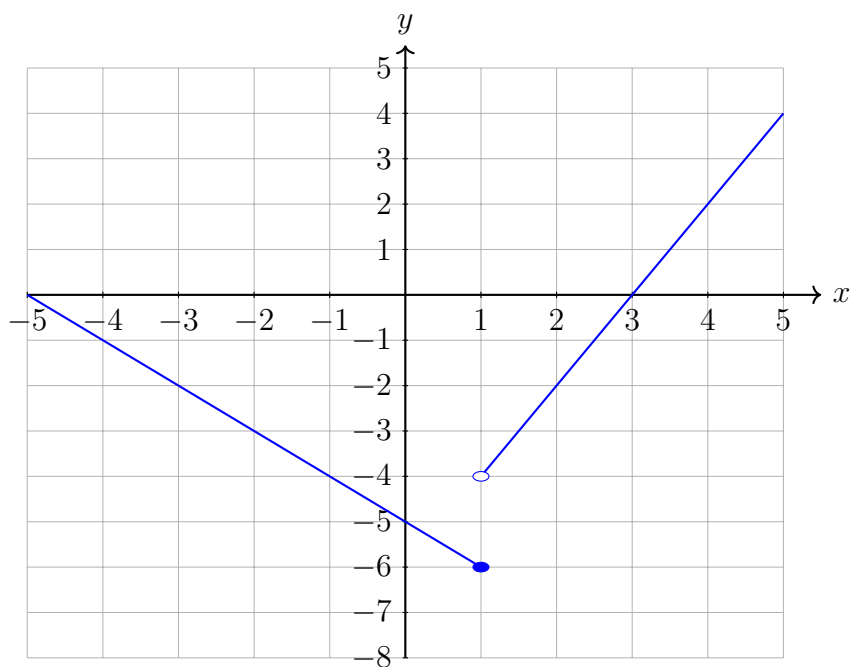


12. The graph of $y = |x|$ is shown below in black. Find an equation for the blue graph that uses the absolute value function.



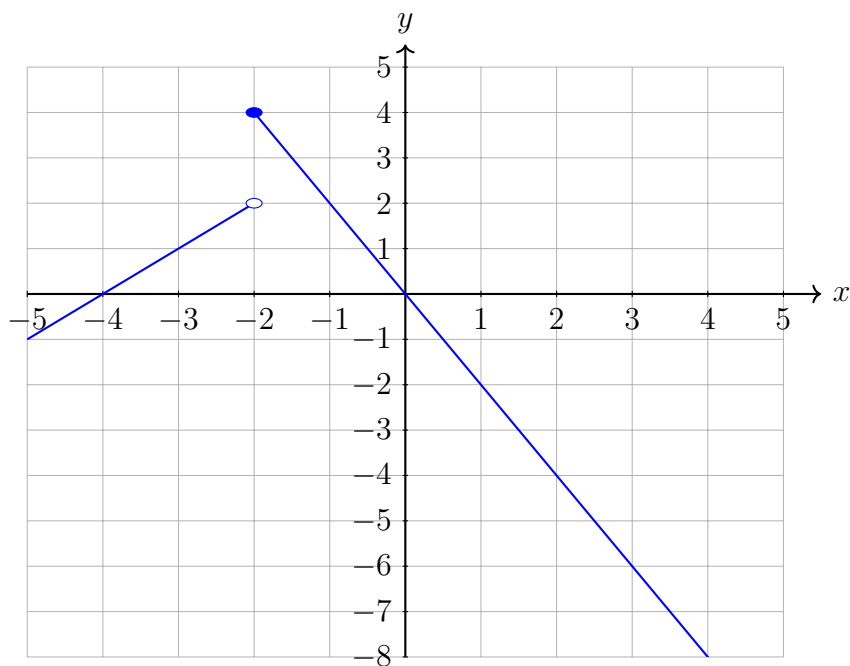
13. Find a piecewise function for the graph below. You may assume that both straight lines continue outside the bounds of the grid shown.

Hint: Using the formula $y - y_1 = m(x - x_1)$, find an equation for each piece of the graph and then write the function as a piecewise function.



14. Find a piecewise function for the graph below. You may assume that both straight lines continue outside the bounds of the grid shown.

Hint: Using the formula $y - y_1 = m(x - x_1)$, find an equation for each piece of the graph and then write the function as a piecewise function.



Questions 15 and 16 investigate drawing graphs of exponential functions and logarithmic functions. While not officially part of Chapter 8 content, it is recommended that you understand the general shape of the graphs of exponential and logarithmic functions.

15. Consider the exponential function $y = e^x$.

- (a) By completing the following table, draw a reasonably accurate graph of $y = e^x$ for $-4 \leq x \leq 3$.

x	-4	-3	-2	-1	0	1	2	3
y								

- (b) Consider that you were now asked to redraw your graph from part (a) but with the domain changed to $-5 \leq x \leq 5$. What would be the minimum and maximum values required for the y -axis to accurately draw this graph?

- (c) State the (natural) domain and range for the function $y = e^x$.

16. Consider the logarithmic function $y = \ln(x)$.

- (a) By completing the following table, draw a reasonably accurate graph of $y = \ln(x)$ for $0 < x \leq 4$.

x	0.01	0.05	0.1	0.25	0.5	1	2	3	4
y									

- (b) In part (a), only positive values of x were considered. Why?

- (c) State the (natural) domain and range for the function $y = \ln(x)$.

17. Consider the function $f(x) = x^2 - 2$ where its domain is all real values of x . Explain why the function $f(x)$ does not have an inverse.

Hint: You may find the horizontal line test helpful. Alternatively, you can show that multiple inputs to the function $f(x)$ result in the same output.

18. (Harder). Show that the inverse of $f(x) = \frac{x+1}{x-1}$ is $f^{-1}(x) = \frac{x+1}{x-1}$. In other words, show that the inverse of $f(x)$ is itself.